UNIVERSITY OF PERADENIYA

DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

END SEMESTER EXAMINATION - SEMESTER I (2022/2023)

CSC1041 - PROGRAMMING LABORATORY I

Answer ALL Questions   Time Allowed: **Three** Hours

PART A - (C PROGRAMMING)

Use the vi editor and save your C program as S21XXXQ1.c where S21XXX stands for your registration number. Extra marks will be awarded for using good programming practices.

1. [50 Marks]

   You are required to create a C program to manage food orders for a restaurant called "Food Fiesta". A recurring menu should be implemented as given below.

   Example Output:

```
Welcome to Food Fiesta!
1. Place an order
2. Print food price list
3. View delivery area details
4. View current orders
5. Clear order records
6. Exit
Please enter your choice:
```

   a. When Option 1 (Place an order) is selected, the program should ask for user data:*Customer Name,Customer ID, Mobile Number* and the user can select the Delivery Area from a predefined list (See the example output below).

   - A customer can order a maximum of 5 different items at a time.
   - After ordering an item, the program should ask if the customer wants to add another item
   - The available food items and their prices are:

Food Items	Price (per item)
Burger	Rs. 150.00
Pizza	Rs. 500.00
Pasta	Rs. 200.00
Salad	Rs. 100.00
Sushi	Rs. 300.00
Fries	Rs. 80.00

- The delivery charges for each destination are:

Destination	Delivery Charge
Colombo	Rs. 100.00
Kandy	Rs. 150.00
Galle	Rs. 200.00
Jaffna	Rs. 250.00
Matara	Rs. 180.00

After placing an order a receipt should be printed with the following details.

- Customer details
- Ordered food items and quantities
- Total order price
- Delivery charge
- Final total topic

Example Output:

```
Please enter your choice: 1
Enter customer name: Taylor
Enter customer ID: 1315V
Enter mobile number: 0713253105

Delivery Areas and Charges:
1. Colombo - Rs. 100.00
2. Kandy - Rs. 150.00
3. Galle - Rs. 200.00
4. Jaffna - Rs. 250.00
5. Matara - Rs. 180.00
Select delivery area: 5

Available food items:
1. Burger - Rs. 150.00
2. Pizza - Rs. 500.00
3. Pasta - Rs. 200.00
4. Salad - Rs. 100.00
5. Sushi - Rs. 300.00
6. Fries - Rs. 80.00

Enter item 1 (1-6): 1
Enter quantity: 1
Do you want to add another item? (y/n): y
Enter item 2 (1-6): 4
Enter quantity: 1
Do you want to add another item? (y/n): n
Order placed successfully! Here is your receipt:

Food Fiesta
Customer Name: Taylor
Customer ID: 1315V
Mobile: 0713253105
Delivery Area: Matara - Rs. 180.00
```

```
Ordered Items:
Burger - 1 - Rs. 150.00 - Total: Rs. 150.00
Salad - 1 - Rs. 100.00 - Total: Rs. 100.00
Total Order Price: Rs. 250.00
Delivery Charge: Rs. 180.00
Final Total Price: Rs. 430.00

Thank you for ordering from Food Fiesta!
```

b. When Option 2 (Print food price list) is selected, the details of all the food items and the price should be displayed as follows.

Example Output:

```
Please enter your choice: 2
Available food items:
1. Burger - Rs. 150.00
2. Pizza - Rs. 500.00
3. Pasta - Rs. 200.00
4. Salad - Rs. 100.00
5. Sushi - Rs. 300.00
6. Fries - Rs. 80.00
```

c. When Option 3 (View delivery area details) is selected, the details of all the delivery areas and delivery charges should be displayed as follows.

Example Output:

```
Please enter your choice: 3
Delivery Areas and Charges:
1. Colombo - Rs. 100.00
2. Kandy - Rs. 150.00
3. Galle - Rs. 200.00
4. Jaffna - Rs. 250.00
5. Matara - Rs. 180.00
```

d. When Option 4 (View current orders) is selected, a list of all current orders should be printed. An example output is given below.

Example Output:

```
Please enter your choice: 4
Current Orders:
Customer Name: Taylor
Customer ID: 1315V
Mobile: 0713253105
Delivery Area: Matara
Ordered Items:
Burger - 1 - Rs. 150.00
Salad - 1 - Rs. 100.00
```

e. When Option 5 (Clear order records) is selected, clear all existing order records, effectively resetting the order system.

Example Output:

```
Please enter your choice: 5
All order records have been cleared.

Welcome to Food Fiesta!
1. Place an order
2. Print food price list
3. View delivery area details
4. View current orders
5. Clear order records
6. Exit
Please enter your choice: 4
No current orders.
```

PART B - (PYTHON PROGRAMMING)

Create a folder named CSC1041_End_S21XXX in your H drive. Save your Python programs inside the same folder as S21XXXQ2.py, S21XXXQ3.py and S21XXXQ4.py for each question where S21XXX is your registration number.

1. [24 Marks]

   a. Write a Python function to pass a filename and a String context as parameters and open the file. (it should print an error message if the file already exists.) Then write the context to the file, and close it.

   b. Using the above function written in (a) open a new file named *context_1.txt* and write the following sentence to it.

      Do not worry, it is easy once you try it!

   c. Then open a new file named *context_2.txt* and write down the following sentence.

      He asked, Is it amazing?

   d. Write a Python function that reads the contents of two text files and prints which file contains the maximum number of characters. Use the above two files to test the program.

   e. Write a Python function that accepts two file names as parameters and append the content from the first file to the second file. Using the function append the content of the *context_1.txt* file to the *context_2.txt* file.

   f. Count the total number of vowels (a, e, i, o, u) in the *context2.txt* file.

   g. Write a Python function to reverse a sentence.

   h. Write a Python function to convert all letters in a sentence to uppercase.

   i. Write a Python function that takes a file name and reads the contents of the file.

      i. First, your program should convert all letters in the file into uppercase letters and then revise the sentence.

ii. Then pass the *context_2.txt* to the function and print the result to the console. (You may use the functions you wrote for (g) and (h)). Your program should handle the exceptions efficiently. The expected outcome is as follows,

```
ITI YRT UOY ECNO YSAE SI TI, YRROW T'NOD?GNIZAMA TI T'NSI" ,DEKSA EH
```

2. [08 Marks]

a. Write a Python program to take a sentence as a user input an replace all 'g' characters with * symbols in the sentence and print it. Give the following sentence as a user input.

google translate

b. Print the following sentence as it is.

"Warm bread's scent; a market's call.
And I love that!"

c. Write a Python script to convert a Celsius value to a Fahrenheit value. Use the following formula.

$$\textit{Fahrenheit} = \textit{Celsius} \times (9/5) + 32$$

The expected outcome is as follows,

```
Enter a sentence: google translate
*oo*le translate

"Warm bread's scent; a market's call.
And I love that !"

Enter the Celsius value: 100
Celsius: 100.0 -> Fahrenheit: 212.0
```

3. [18 Marks]

a. Write a Python function to get 3 user inputs and store them using a suitable collection type. Make sure the user is inserting numeric data. Write code to properly handle invalid data types.

b. Write a Python function to calculate and return the sum of 3 numbers. You have to pass the above-decided collection type in part (a) as a parameter.

c. Write a Python function to calculate and return the average of 3. You have to pass the above-decided collection type in part (a) as a parameter.

d. A teacher wanted to store the following data of two terms of 3 students using a simple Python program.

Semester 1	Semester 2
90	30
100	91
79	55

With the support of the above three functions in (b) and (c) calculate the,

i. The class sum of each semester

ii. The Class average for each semester

iii. The Class average for both semesters

iv. The Class average for both semesters

The expected outcome is as follows,

```
Enter the marks for semester 1
Enter a number: 90
Enter a number: 100
Enter a number: 79

Enter the marks for semester 2
Enter a number: 30
Enter a number: 91
Enter a number: 55

The class sum for semester 1 is:269.0
The class sum for semester 2 is:176.0
The class sum for both semesters:445.0

The class average for semester 1 is:89.66666666666667
The class average for semester 2 is:58.666666666666664
The class average for both semesters:74.16666666666667
```

—End of the Paper—